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EXPLOITING INATE MOTOR DYNAMICS FOR HIGH STIFFNESS HAPTIC DISPLAY ON BRUSHLESS DC MOTORS

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ABSTRACT

The maximum stiffness that can be rendered by impedance-controlled haptic devices has traditionally been limited by quantization, discretization, and delay in the digital loop. Recent research has shown that performance can be improved by utilizing the natural inductive stiffness inherent in brushed DC motors. This study extends the concept of exploiting a motor's dynamics to the three phase brushless DC (BLDC) motor. It is analytically shown that the inductances of a BLDC motor's windings map to a high physical stiffness. This stiffness is made available at frequencies important to haptic interaction by cancelling the resistance in each winding with analog feedback, effectively slowing the motor's electrical dynamics. Experimental verification is obtained by implementing the proposed spring drive in analog circuitry in combination with a digital position feedback loop. The final results support the analytic solution and compare the spring drive favorably with traditional current and voltage drives for haptic applications.

INTRODUCTION

The effective rendition of interactions ranging from free motion to rigid contact with a virtual surface is a central goal in haptic simulation. Impedance-type haptic devices try to accomplish this by producing forces in response to contacts in the virtual environment as a result of motion caused by the user. In the case of free motion, these devices work very well, since they are generally designed to be easily back-driven by the user. Stable simulation of high-stiffness contact, on the other hand, remains a challenge. Limits on maximum stiffness are imposed by computational delays [1], time discritization [2] [3], and position quantization [4] [5], all of which lead to instability of the digital position control loop. Realistic stiff contact is further complicated by the need for stiffness over the wide band of frequencies at which humans perceive haptic stimuli (up to 1 kHz [6]).

Traditional impedance devices typically rely on small DC motors to deliver forces to the user. Most often, these motors are driven by current amplifiers that speed up and hide the electrical dynamics. Recent work has shown, however, that new motor drive and control schemes implemented in analog electronics can significantly improve the stability and stiffness of impedance devices. Diolaiti and Niemeyer propose in [7] that hiding the electrical dynamics with current amplifiers overlooks the high natural stiffness included in the inductance of the DC motor. They exploit this natural stiffness, which is available and stable at all frequencies but often ignored due to the dominant resistive damping at low frequencies. Their approach is accomplished by cancellation of the the motor's resistance with an analog current feedback controller. Another approach is to increase the physical damping in the haptic display through active electrical damping in analog electronics [8]. Similarly, [9] uses the electrical dynamics to increase viscous friction and improve stability.

In this paper the concept of utilizing the natural inductive stiffness introduced in [7] is extended to brushless DC (BLDC) motors. This is motivated by the existence of electrical discontinuities in brushed motors caused by mechanical commutation. Since utilizing the inductive stiffness at low frequencies relies on the cancellation of motor resistance, discontinuity-free BLDC motors are advantageous. Additionally, BLDC motors are superior to their brushed counterparts in terms of power density, reliability, rotor inertia, efficiency, and back-drivability.

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In order to extend this concept to the BLDC case, an analysis of the motor dynamics is performed, showing that an inductive stiffness of $K_L = \frac{3k_T^2}{2L}$ exists in the mechanical domain. A new type of analog BLDC motor drive is designed to take advantage of this stiffness for rendering stiff contact. Further, to interface this new drive to a virtual environment, integration of a digital loop is described. The system is implemented alongside traditional voltage and current drives and demonstrates significant stiffness improvements.

The paper is structured as follows. Section 2 reviews the electrical dynamics of typical brushed motors and describes the concept of utilizing the motor dynamics to provide a high natural stiffness. Section 3 covers the operation and dynamics of brushless DC motors, and the ideas of section 2 are extended to the brushless case in section 4 through the analytical development of a motor resistance canceling controller. In section 5 a brushless motor driver that implements this controller is proposed, implementation details are discussed, and this new 'spring' drive is compared to traditional voltage and current amplifier configurations. Finally, section 6 presents experimental results verifying the expected operation of the spring drive and comparing performance to traditional methods.

PHYSICAL INTERPRETATION OF MOTOR DYNAMICS

Previous work has shown that the maximum stiffness of a haptic display can be improved significantly by exploiting the natural dynamics of DC motors [7]. This is accomplished by replacing the traditional current amplifier with an analog circuit that couples the user to the virtual environment through the natural inductive stiffness of the motor. The concept is described here in the brushed DC motor case as a foundation for its extension to brushless DC motors in section 4.

The electrical dynamics of a typical brushed DC motor are given by,

$$e_A(t) = Ri(t) + L\frac{di(t)}{dt} + e_B(t) \tag{1}$$

$$e_B(t) = k_T\dot{\theta}(t) \tag{2}$$

where e_A is the applied voltage, e_B is the back-EMF, k_T is the motor constant, and R and L are the winding resistance and inductance, respectively. The mechanical dynamics are,

$$m\ddot{\theta}(t) = F(t) - c(\dot{\theta}(t)) - F_H(t) \tag{3}$$

$$F(t) = k_T i(t) \tag{4}$$

where m is the rotor inertia, $c(\dot{\theta})$ is the friction torque, and F_H is an input torque opposing the velocity of the rotor. The coupling equations Eq. (2) and Eq. (4) relate torque to current and voltage to velocity, interfacing the electrical and mechanical dynamics. As a result, it is possible to to interpret the electrical dynamics in the mechanical domain. The inductor, an energy storage element, is transformed into an equivalent mechanical spring,

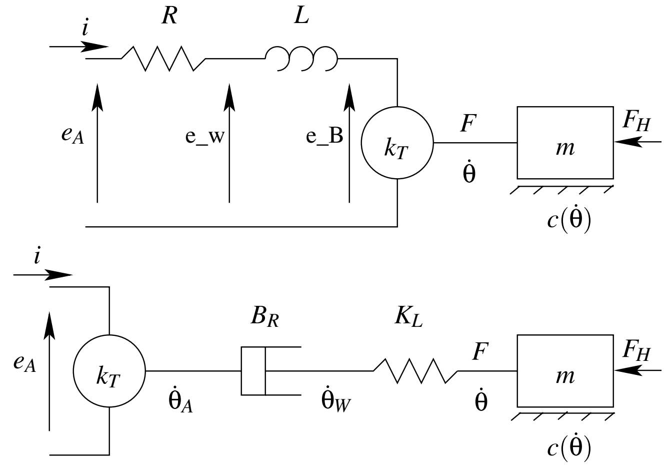

Figure 1. MOTOR INDUCTANCE AND RESISTANCE MAPPED TO THE MECHANICAL DOMAIN

$$K_L = \frac{k_T^2}{L} \tag{5}$$

The resistor, an energy dissipating element, is transformed into an equivalent mechanical viscous damper,

$$B_R = \frac{k_T^2}{R} \tag{6}$$

In the mechanical domain these equivalent elements are connected in series. Figure 1 illustrates this mapping of L and R to the mechanical domain. It is noted in [7] that motors typically used in haptics have low inductance and, therefore, high stiffness K_L. In the case of the Maxon RE 25, it was found that K_L, at 2 Nm/rad, is almost seven times stiffer than the best stable virtual spring implemented with traditional methods.

With the goal of accurately rendering stiff contact in haptic simulation, it becomes attractive to try to couple a user to the virtual environment through K_L. This inductive spring is included in the motor's dynamics, inherently stable, available at all frequencies, and typically very stiff. However, since K_L is connected in series with B_R, the behavior of the interface is dominated by the damper over the relatively low frequencies of importance during the interaction. In order to take advantage of the motor's natural stiffness for haptics it is necessary to reconsider the traditional current drivers used as motor amplifiers.

As seen in [7] a simple controller can be implemented to lock the damper B_R by canceling the motor's resistance. By interpreting only L in the mechanical domain, the electrical dynamics simplify to,

$$e_W = e_A - Ri \qquad\qquad \dot{\theta}_W = \frac{e_W}{k_T} \tag{7}$$

Therefore, regulating the velocity of the end point of K_L, $\dot{\theta}_W$,to zero and locking the damper can be accomplished by setting,

$$e_A = Ri \tag{8}$$

This voltage controller compensates for the motor's resistance R and regulates $e_W = 0$ and $\dot{\theta}_W = 0$. It can be implemented in analog circuitry by a simple voltage drive with positive feedback of Ri as a replacement for the typical current driver. Unlike a current driver, which speeds up and hides the electrical dynamics, this new amplifier slows down the natural electrical response and leverages the advantageous physical properties.

BRUSHLESS DC MOTOR

The controller described above utilizes a single-value model of R. Brushed DC motors, however, experience resistance discontinuities due to commutation, and this can cause discontinuities in the force experienced by the user during stiff contact as demonstrated in [7]. This study extends the idea of utilizing the natural motor dynamics through resistance compensation to brushless DC (BLDC) motors, in part to address this issue. More generally, however, BLDC motors out-perform brushed DC motors, exhibiting higher power density, reliability, and efficiency. They also have lower rotor inertia and torque ripple (with sinusoidal commutation). Consequently, BLDC motors have been used in several high performance haptic device designs. This section will review the operation and dynamics of BLDC motors before analysis of BLDC resistance compensation in the next section.

Brushless DC motors are constructed with the permanent magnet located on the rotor and several (usually three) windings on the stator. By moving the permanent magnet to the rotor, there is no need to supply current to the rotating element, and brushes or slip rings are unnecessary. In the case of the three phase BLDC motor, the windings are each separated by an angle of 120 degrees. Each winding experiences a sinusoidal back-EMF voltage. External circuitry and control is necessary to commutate a BLDC motor, and several methods are available, most notably six-step block commutation and sinusoidal commutation. The former is inexpensive, requiring only three Hall sensors, which are usually integrated into the motor. Sinusoidal commutation requires position feedback via an encoder or resolver, but theoretically has no torque ripple and can be used for positioning applications. Additional details can be found in [10].

The electrical dynamics of a three-phase BLDC are,

$$e_{A1}(t) - e_n(t) = Ri_1(t) + L\frac{di_1(t)}{dt} + e_{b_1} \tag{9}$$

$$e_{A2}(t) - e_n(t) = Ri_2(t) + L\frac{di_2(t)}{dt} + e_{b_2} \tag{10}$$

$$e_{A3}(t) - e_n(t) = Ri_3(t) + L\frac{di_3(t)}{dt} + e_{b_3} \tag{11}$$

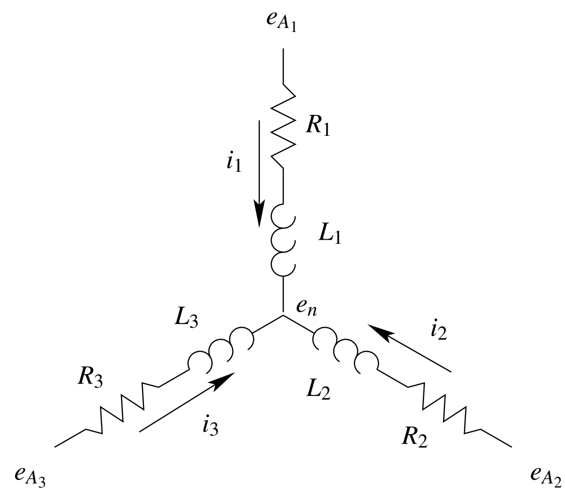

Figure 2. 3-PHASE BLDC MOTOR WINDINGS IN WYE CONFIGURATION

with the back-EMF voltages,

$$e_{b_1} = k_T\dot{\theta}\sin(\theta) \tag{12}$$

$$e_{b_2} = k_T\dot{\theta}\sin(\theta - 120^\circ) \tag{13}$$

$$e_{b_3} = k_T\dot{\theta}\sin(\theta + 120^\circ) \tag{14}$$

Note that by definition the three currents add to zero

$$i_1 + i_2 + i_3 = 0 \tag{15}$$

and each winding contributes a force in a sinusoidal fashion

$$F_1 = k_T i_1 \sin(\theta) \tag{16}$$

$$F_2 = k_T i_2 \sin(\theta - 120^\circ) \tag{17}$$

$$F_3 = k_T i_2 \sin(\theta + 120^\circ) \tag{18}$$

with the total force

$$F = F_1 + F_2 + F_3. \tag{19}$$

Figure 2 shows the corresponding BLDC windings in a wye configuration. The less common, but analytically identical delta configuration is not considered for this analysis.

For this study sinusoidal commutation was used to provide constant torque versus the rotor angle and allow for position control. This means that for a given current command, i_c, the commutated current commands to each winding are,

$$i_{c_1} = i_c \sin(\theta) \tag{20}$$

$$i_{c_2} = i_c \sin(\theta - 120^\circ) \tag{21}$$

$$i_{c_3} = i_c \sin(\theta + 120^\circ) \tag{22}$$

Assuming current tracking, substituting Eqs. (20-22) into Eqs. (16-18,19) and simplifying yields an expression for the equivalent torque during sinusoidal commutation,

$$\begin{aligned} F &= k_T i_c[\sin^2(\theta)+\sin^2(\theta-120^\circ)+\sin^2(\theta+120^\circ)] \\ &= \frac{3}{2}k_T i_c \end{aligned} \tag{23}$$

EXPLOITING BLDC MOTOR DYNAMICS

Utilizing the natural dynamics of a BLDC motor to couple a user to the virtual environment is conceptually identical to the approach taken in section 2. However, now the resistances of three separate windings must be canceled by a trio of compensators. For the following analysis an ideal case is considered where the resistances and inductances of each of the three windings are identical and equal to R and L, respectively. As a consequence, by adding Eqs. (9-11) the common node voltage is the mean of the three terminal voltages,

$$e_n = \frac{e_{A1}+e_{A2}+e_{A3}}{3} \tag{24}$$

All three drivers cancel the resistance

$$e_{A1}(t) = Ri_1(t) \tag{25}$$

$$e_{A2}(t) = Ri_2(t) \tag{26}$$

$$e_{A3}(t) = Ri_3(t) \tag{27}$$

such that from Eq. (24) and Eq. (15) the common node voltage $e_n = 0$. Substituting Eqs. (25-27) into Eqs. (9-11) yields,

$$\frac{di_1}{dt} = -\frac{k_T}{L}\dot{\theta}\sin(\theta) \tag{28}$$

$$\frac{di_2}{dt} = -\frac{k_T}{L}\dot{\theta}\sin(\theta-120^\circ) \tag{29}$$

$$\frac{di_3}{dt} = -\frac{k_T}{L}\dot{\theta}\sin(\theta+120^\circ) \tag{30}$$

Integrating and substituting each i into Eqs. (16-18,19) provides the result,

$$\begin{aligned} F(t) = -\frac{k_T^2}{L}[&\cos(\theta_0)\sin(\theta) \\ &+\cos(\theta_0-120^\circ)\sin(\theta-120^\circ) \\ &+\cos(\theta_0+120^\circ)\sin(\theta+120^\circ)] \end{aligned} \tag{31}$$

where $\theta_0 = \theta(0)$ with $F(0) = 0$. This simplifies to,

$$F(t) = -\frac{3k_T^2}{2L}\sin(\theta-\theta_0) \tag{32}$$

Thus, over small angles the equivalent inductive spring for a BLDC motor is

$$K_L = \frac{3k_T^2}{2L} \tag{33}$$

In the case where the current feedback gain of the controller is not exactly R, a small resistance dR remains uncanceled leading to an equivalent resistive damper:

$$B_R = \frac{3k_T^2}{2dR} \tag{34}$$

It is important to note that B_R causes drift at low frequencies, while K_L dominates at high frequencies. The cutoff frequency between the two effects is $\omega_{\text{cutoff}} = \frac{dR}{L}$. For the remaining discussion, 'high frequency' and 'low frequency' refer to the frequency bands corresponding to the effects above and below this cutoff, respectively.

IMPLEMENTATION OF A SPRING DRIVE

The controller described above, which will be referred to as a spring drive, is implemented in analog circuitry similar to two common motor amplifier configurations, the voltage drive and current drive. Recall that a voltage drive simply achieves a command voltage at the load. A current drive feeds back the load current to track a command current, often with an integral term to compensate for back-EMF. The proposed spring drive uses positive feedback of Ri as detailed previously to make K_L visible. In order to compare the maximum achievable stiffnesses of each approach, all three drivers have been implemented. Additionally, to perform absolute position feedback from encoder measurements, a digital PD loop has been included in each case. This PD controller provides a command signal for the rendering of virtual walls. Equations (35-37) show expressions for e_A, the applied voltage, for the voltage drive, current drive, and spring drive, respectively.

$$e_{A_j} = e_{B_j} + Ri_{c_j} \tag{35}$$

$$e_{A_j} = e_{B_j} + Ri_{c_j} + K_f(i_{c_j} - i_j) \tag{36}$$

$$e_{A_j} = \hat{R}i_j + \hat{dR}i_{c_j} \tag{37}$$

Note that for each type of drive, three separate driver circuits are necessary, with each driving a single winding ($j = 1,2,3$). The terms $\hat{R}$ and $\hat{dR}$ are respective approximates of the winding resistance R and the residual resistance dR remaining due to inexact cancellation of R in the spring drive. The system details giving rise to these equations are described below.

Figure 3 shows a block diagram of the combined analog and digital system, illustrating the connection between the PD position controller, analog drivers, and physical plant. The motor

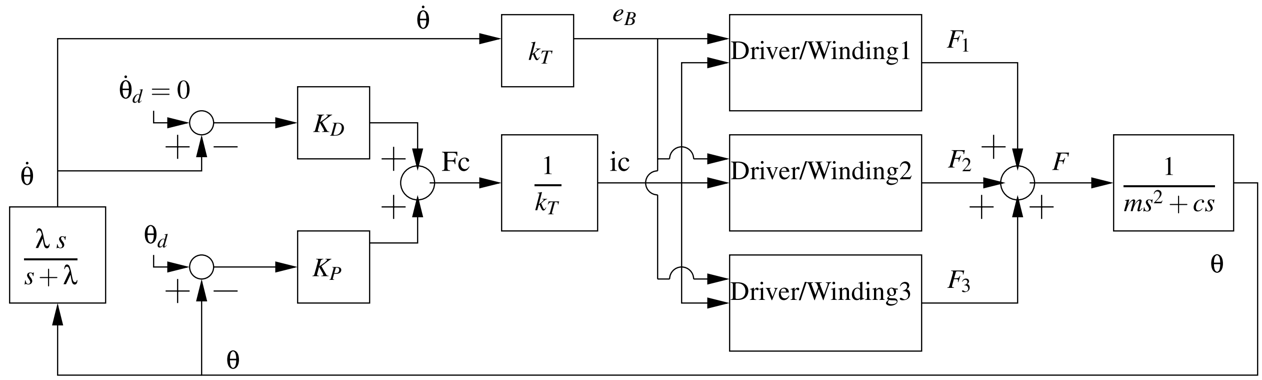

Figure 3. SYSTEM BLOCK DIAGRAM OF THE DIGITAL PD LOOP FOR CURRENT, VOLTAGE, OR SPRING BLDC MOTOR DRIVERS.

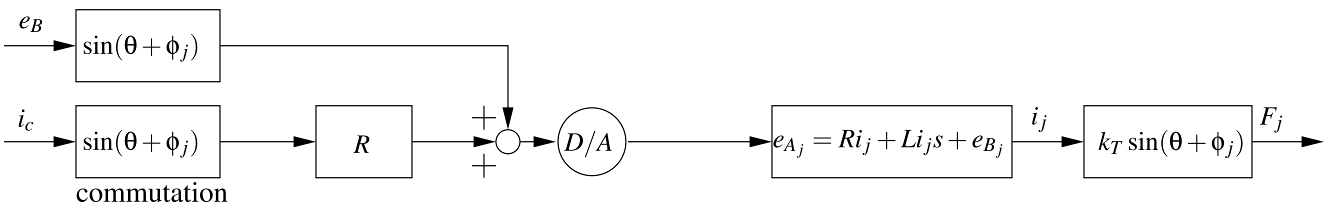

Figure 4. VOLTAGE DRIVE FOR ONE WINDING

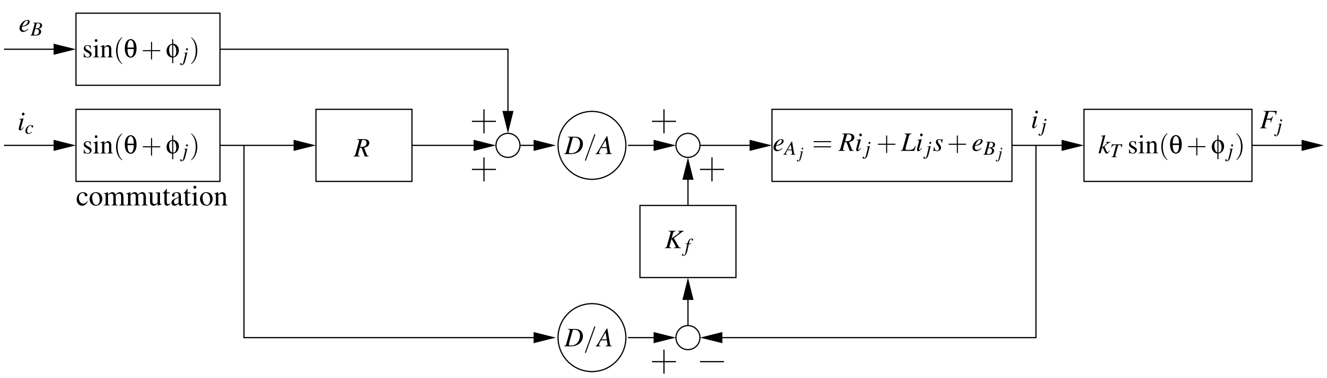

Figure 5. CURRENT DRIVE FOR ONE WINDING

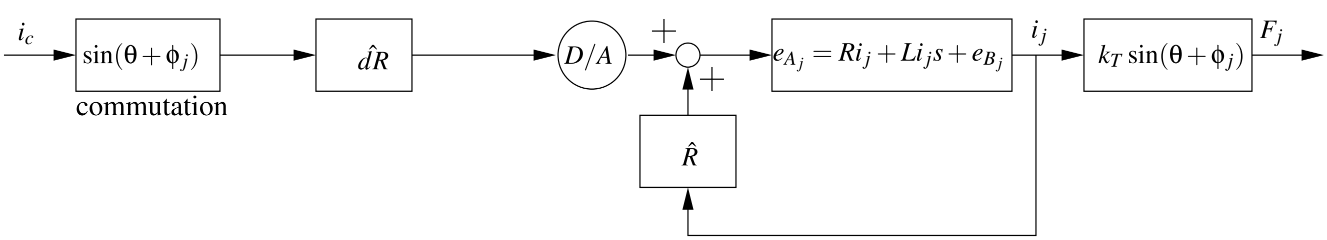

Figure 6. SPRING DRIVE FOR ONE WINDING

driver details have been black-boxed in this figure, however, and Figures 4, 5, and 6 feature separate diagrams of the voltage, current, and spring drives. As with the equations above, each of these three figures details the implementation for a single winding.

The system's digital loop counters movement away from a desired set-point by using a rotary encoder to obtain absolute angle measurements. A restoring force command F_c is generated by PD position feedback using these measurements and a filtered velocity estimate,

$$F_c = K_P(\theta_d - \theta) - K_D\dot{\theta} \tag{38}$$

This expression represents a digital spring-damper that produces a DC force under constant deflection. A command current is obtained by Eq. (4) and mapped to a voltage signal through a gain specific to each driver type (see below), such that the steady state force produced equals F_c. The velocity estimate can explicitly cancel back-EMF to simplify the driver circuits.

In the voltage drive configuration, the motor current is generated open loop. The commutated command currents are multiplied with R to provide the open loop voltage command that should produce F_c. This type of drive operates with the motor's natural bandwidth of 1 kHz. Explicit back-EMF compensation is added to the applied voltage.

The current drive adds a negative current feedback loop with a feedback gain K_f. A gain of $K_f = 10$ was selected experimentally to provide good current tracking, accelerating the electrical dynamics more than 30-fold to above 30 kHz. Again, explicit back-EMF compensation is used, eliminating the need for a lag compensator.

The spring drive is conceptually different. Much of the restoring force, especially at high frequency, is generated in analog electronics. Therefore, the digital force command is only responsible for canceling low frequency drift due to any residual dR, through which the command currents are mapped to generate a voltage signal. Practically, $\hat{R}$ is set as a best estimate, recognizing that the real R does fluctuate. The residual estimate $\hat{dR}$ is then set experimentally such that the programmed K_P and the measured DC stiffness agree. Uncertainties in dR mean that the actual DC stiffness may differ from the programmed value, though this does not affect the maximum achievable K_P. The spring drive implicity utilizes the back-EMF effect and does not include compensation.

EXPERIMENTAL RESULTS AND DISCUSSION

Evaluation of the proposed BLDC motor spring drive is performed, and Table 1 lists parameter values for the actual experimental setup. Here, the BLDC motor used is a Maxon EC32, and the system elements impacting digital loop performance include a 1024 line incremental encoder with 4X quadrature and a computer servo rate of 3 kHz. During experiments input torques are applied directly to the motor shaft, and the digital control loop implements a two sided wall. The absolute position of this wall is held constant for all tests to avoid any potential dependenies on the rotor angle.

R_{motor}	0.3Ω	$\hat{R}$	0.324 Ω
R_{sense}	0.05Ω	f_{servo}	3 kHz
L	56.7 μH	λ	180 Hz
K_T	4.93 mNm/A	Encoder	4096 CPR
K_f	10 Ω		

Table 1. PARAMETER VALUES

Two experiments are performed. In the first experiment, the spring drive and voltage drive are both tuned to the same digital PD stiffness, K_p, and their actual stiffnesses are measured in response to high and low frequency torque inputs. Once again, 'high' and 'low' frequency as used here are relative to the cutoff $\omega_{\text{cutoff}} = \frac{dR}{L}$, and correspond to approximate impulse and DC inputs, respectively. The goal is to show that for low frequency inputs, where the residual resistance dR causes drift, the stiffness of both amplifier configurations is identical, namely K_p. Conversely, for high frequency inputs, it is expected that the spring drive will exhibit the motor's natural stiffness K_L, while the stiffness of the voltage drive will remain at K_p. For this experiment K_p is selected as the maximum achievable PD stiffness of the voltage drive before single encoder count limit cycling occurs at the digitally implemented two-sided wall. The second experiment is a performance comparison between all three drive types. Each controller is tuned to its maximum digital loop stiffness as defined above and subjected to high and low frequency torque inputs.

Figure 7 shows the results of several low frequency torque inputs against the virtual wall for the spring and voltage drives in the first experiment. Both torque versus rotor angle curves have the programmed stiffness, K_p, as expected, where $K_p = 100$ mNm/rad. For high frequency inputs the stiffness of the voltage drive remains near K_p, while the spring drive produces a value of 654 mNm/rad (Fig. 8). The expected value of K_L from Eq. (33) and the measured motor parameters is

	Programmed	Measured	
Drive Type	Digital K_p	Low f Input	High f Input
Voltage	100 mNm/rad	96 mNm/rad	150 mNm/rad
Current	120 mNm/rad	123 mNm/rad	244 mNm/rad
Spring	475 mNm/rad	479 mNm/rad	698 mNm/rad

Table 2. MAXIMUM STIFFNESS ACHIEVED FOR LOW AND HIGH FREQUENCY INPUTS

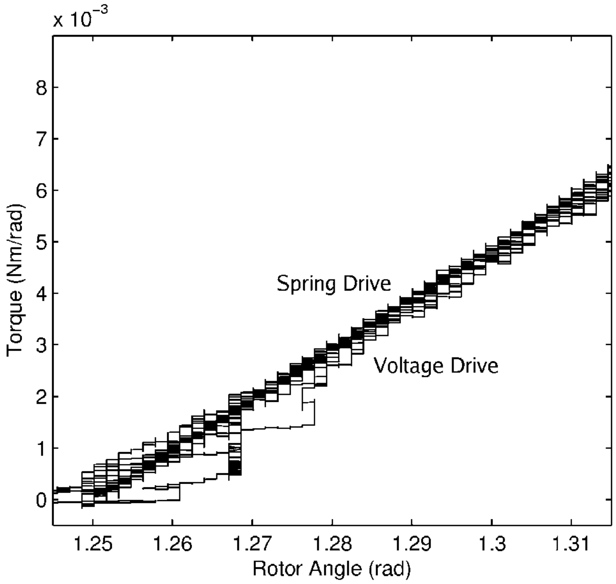

Figure 7. EXPERIMENT 1: STIFFNESS OF SPRING AND VOLTAGE DRIVE IN RESPONSE TO A LOW FREQUENCY INPUT. BOTH DRIVES IMPLEMENT THE SAME DIGITAL PD CONTROLLER. LOW FREQUENCY STIFFNESSES ARE INDISTINGUISHABLE

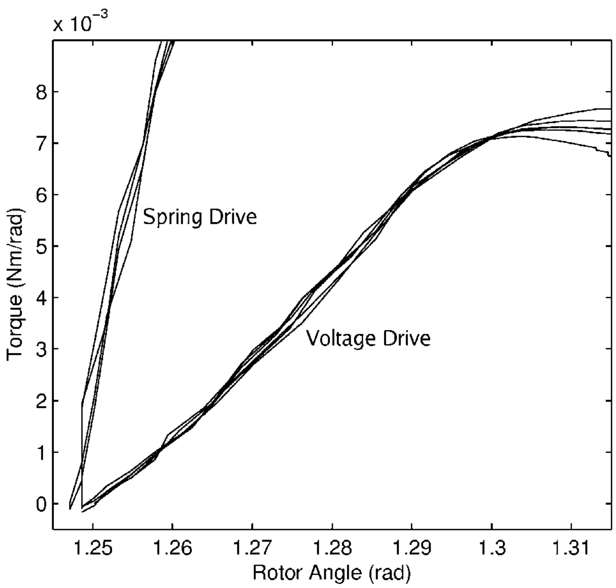

Figure 8. EXPERIMENT 1: STIFFNESS OF SPRING AND VOLTAGE DRIVE IN RESPONSE TO A HIGH FREQUENCY INPUT. BOTH DRIVES IMPLEMENT THE SAME DIGITAL PD CONTROLLER

$K_L = 644$ mNm/rad, and the results agree well with the analytic result. Note that the quantization in the plots is a limitation of the encoder resolution for these small deflections.

For the motor drive performance comparison in experiment

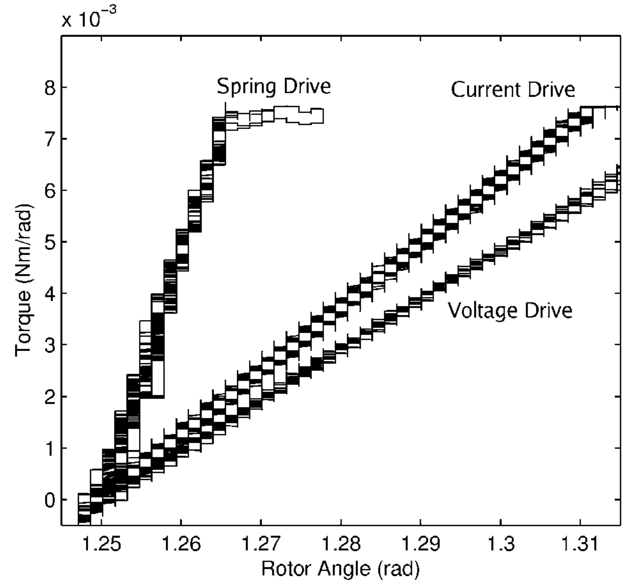

Figure 9. EXPERIMENT 2: COMPARISON OF MAXIMUM STIFFNESS FOR VOLTAGE, CURRENT, AND SPRING DRIVES IN RESPONSE TO A LOW FREQUENCY INPUT

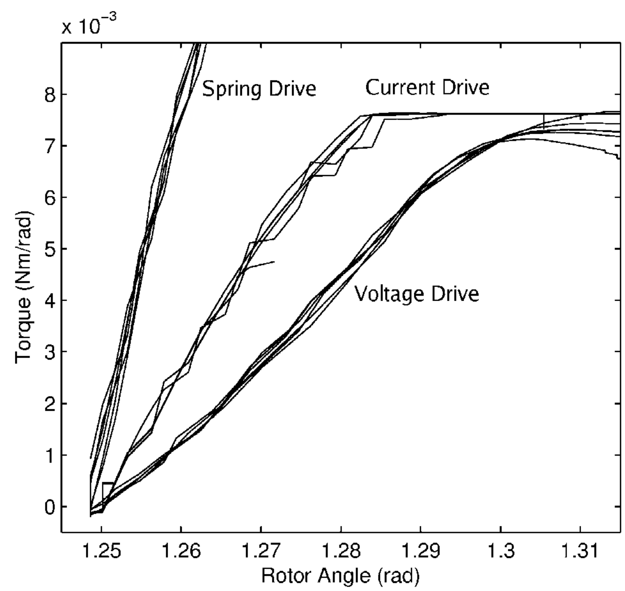

Figure 10. EXPERIMENT 2: COMPARISON OF MAXIMUM STIFFNESS FOR VOLTAGE, CURRENT, AND SPRING DRIVES IN RESPONSE TO A HIGH FREQUENCY INPUT

two, the digital PD controller is tuned for a maximum stable stiffness, K_p, for each driver. Table 2 shows these programmed values as well as the measured stiffnesses in response to high and low frequency input torques for each driver. The torque versus rotor angle curves are displayed in Fig. 9 for the low frequency inputs and Fig. 10 for the high frequency inputs. The interesting result is that the spring drive, in addition to its high frequency K_L stiffness, is capable of much higher digital loop stiffness than the voltage and current drives. This effect may be a result of the natural stiffness reducing the high frequency load on the PD controller. Now relegated to compensating for the low frequency, B_R-induced drift, the PD controller is less sensitive to performance limits imposed by quantization, discritization, and time delay, resulting in improved maximum digital stiffness.

CONCLUSION

We showed both analytically and experimentally that, as with brushed motors, the inductances of a BLDC motor's windings map to a high physical stiffness $K_L = \frac{3k_T^2}{2L}$. The experiment demonstrated that the proposed spring motor driver configuration is capable of producing this stiffness at high frequencies, exceeding the maximum stable stiffness achieved with current and voltage drives by factors of 2.9 and 3.7, respectively. Additionally, it was shown that the spring drive is capable of higher stable digital PD gains at low frequencies, producing stiffnesses at least 3.9 times higher than the traditional drivers. Improving the stiffness of haptic devices, especially at high frequency, is important as the transient signals during virtual contact relay significant information to the user.

Practically, the spring drive implements a physical coupling between the haptic device and the virtual world. Consequently, the input to the spring drive is fundamentally a voltage describing the motion of the set-point for this coupling. This converts the controller into an admittance element producing motion instead of force, which underlines the improvements to creating rigid contacts. Future work will utilize this further and enable arbitrary multi-degree-of-freedom environments to be described in such a framework.

Also, future work needs to address any potential imbalance of resistances or current feedback gains in the three windings. Our assumption of symmetry worked well for small deflections, but may cause changes in stiffness over the rotor angle.

Considering that BLDC motor windings are available with inductive stiffnesses several times larger than the motor implemented for this study, the potential for performance improvement through use of the spring drive is substantial. We hope that the gains provided by the spring approach, coupled with the high performance qualities of BLDC motors, will prove useful for the design of very stiff impedance-type haptic devices.

ACKNOWLEDGMENT

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